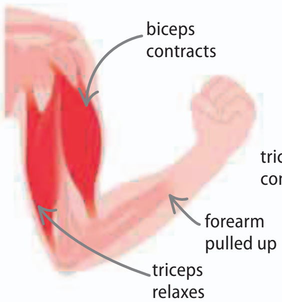
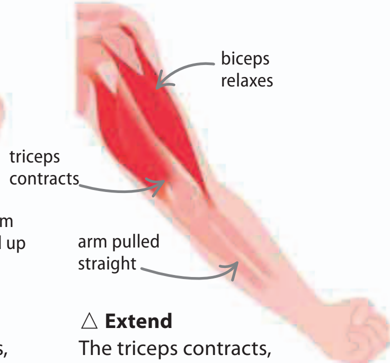
Anchor points

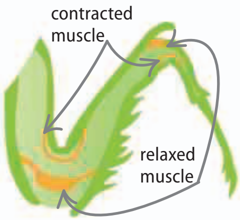
Muscles exert a force by contracting, or pulling, and need a solid anchor point to pull against. This is the main function of a skeleton, with the bones connecting at joints, to allow it to move when muscles pull. Muscles cannot push, so they work in pairs, with each muscle pulling in the opposite direction to the other.



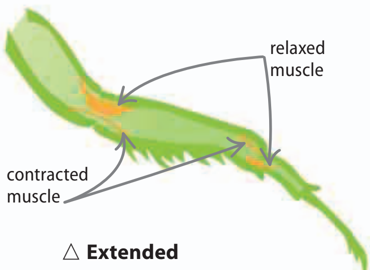
△ **Flex**
The biceps muscle contracts, pulling up on the forearm and causing the whole arm to bend at the elbow.



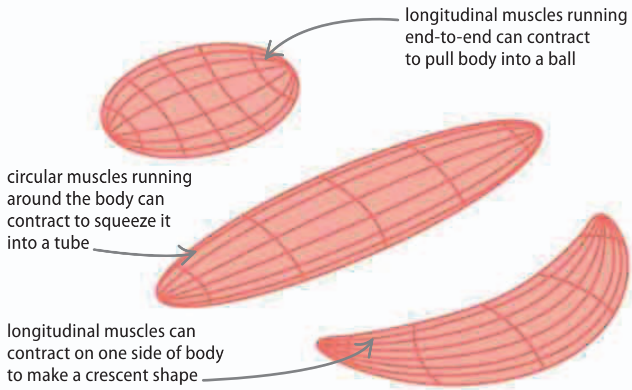
△ **Extend**
The triceps contracts, and the biceps relaxes, pulling the forearm down and straightening the arm.



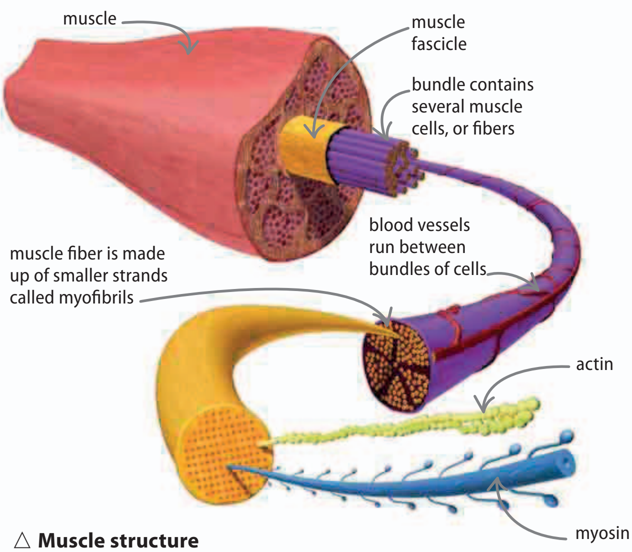
△ **Raised**
Arthropod exoskeletons contain pairs of muscles attached to their jointed inside surfaces.



△ **Extended**
The exoskeleton does not bend when pulled by a muscle. Instead, the force is transferred to the joint, making the whole joint move.



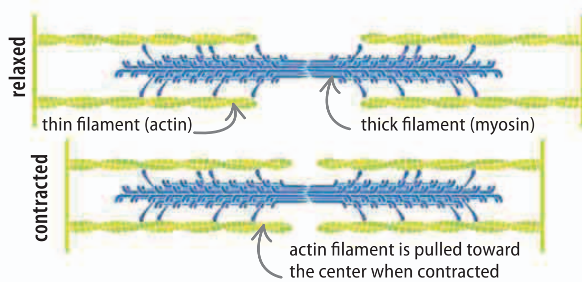
△ **Hydrostatic skeleton**
Worms and other soft-bodied animals have a hydrostatic skeleton—made of sacs of liquid surrounded by muscles. These have a fixed volume, but can be changed into different shapes using sets of circular and longitudinal muscles.



△ **Muscle structure**
Muscles are formed from a hierarchy of bundles. Even the smallest muscle contains several fascicles, which are bundles of muscle cells. In turn, the cells contain bundles of myofibrils that are filled with myosin and actin.

Muscle contraction

A muscle cell takes the form of a long fiber—up to 30 cm (12 in) long in a man’s thigh. The cell contains many hundreds of nuclei and several bundles of myofibrils, which are made up of two protein filaments known as myosin and actin. Muscles contract when the two filaments move closer together in the cells. Millions of these tiny movements accumulate into a powerful contraction.



△ **Actin and myosin**
When a muscle receives an electric pulse from a nerve, the signal causes the thick myosin protein to haul itself along two actin strands, pulling them toward the center. When relaxed, the proteins spread apart again, and the muscle lengthens.